

Session 10: StringsTopics:Declaring and handling strings.Common functions: strlen(), strcpy(), strcmp().Example: Create a username from name input.

Tasks

1.

Declare a string variable called songTitle and assign it the value 'Tum Hi Ho'. Print the length of the string using strlen().

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

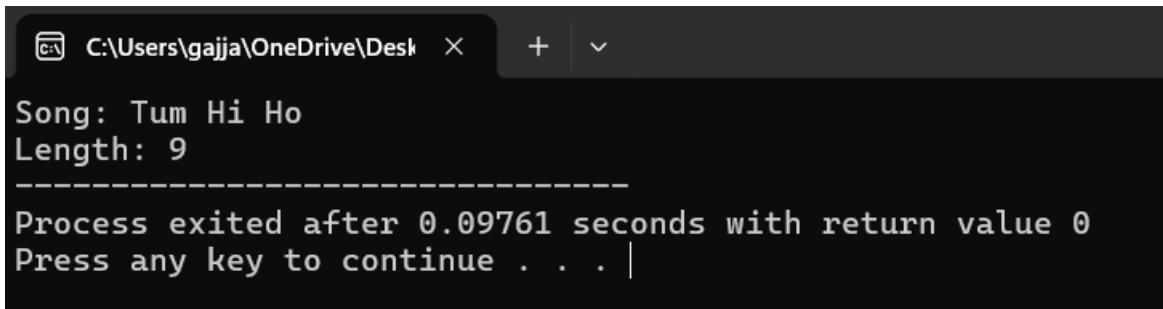
```
    char songTitle[] = "Tum Hi Ho";
```

```
    printf("Song: %s\n", songTitle);
```

```
    printf("Length: %lu", strlen(songTitle));
```

```
    return 0;
```

```
}
```



```
C:\Users\gajja\OneDrive\Desktop  X  +  v
Song: Tum Hi Ho
Length: 9
-----
Process exited after 0.09761 seconds with return value 0
Press any key to continue . . . |
```

2.

Take input for two usernames (as strings) and compare them using strcmp(). Display whether they are the same or different.

```
#include <stdio.h>
```

```
#include <string.h>
```

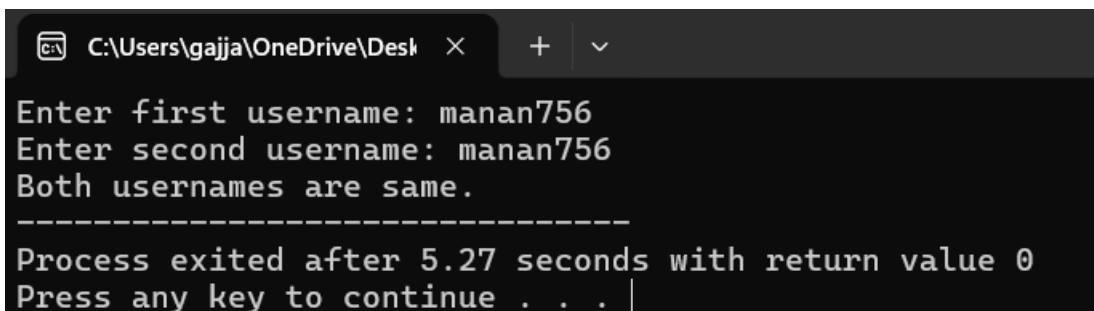
```
int main()
{
    char username1[50];
    char username2[50];

    printf("Enter first username: ");
    scanf("%49s", username1);

    printf("Enter second username: ");
    scanf("%49s", username2);

    if(strcmp(username1, username2) == 0)
    {
        printf("Both usernames are same.");
    }
    else
    {
        printf("Usernames are different.");
    }

    return 0;
}
```



```
C:\Users\gajja\OneDrive\Desk  X  +  v
Enter first username: manan756
Enter second username: manan756
Both usernames are same.
-----
Process exited after 5.27 seconds with return value 0
Press any key to continue . . . |
```

3.

Copy the string 'Flipkart' into another string variable called shoppingApp using strcpy(), then print the value of shoppingApp.

Hint: Make sure to declare enough space for the destination string.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

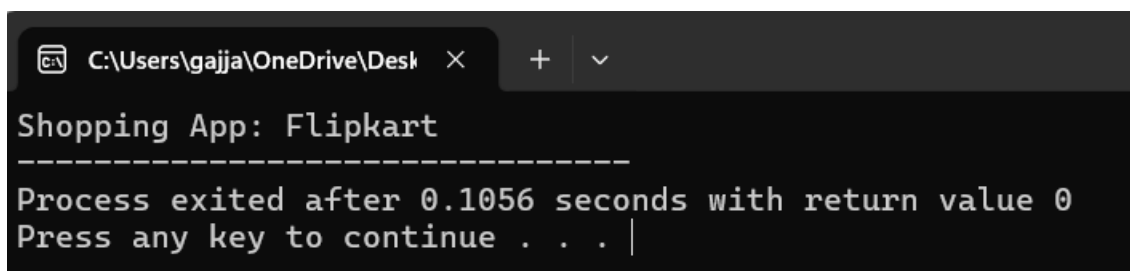
```
    char shoppingApp[20];
```

```
    strcpy(shoppingApp, "Flipkart");
```

```
    printf("Shopping App: %s", shoppingApp);
```

```
    return 0;
```

```
}
```

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\gajja\OneDrive\Desktop' and standard window controls. The command prompt displays the output of the program: 'Shopping App: Flipkart' followed by a dashed line separator. Below the separator, it shows 'Process exited after 0.1056 seconds with return value 0' and 'Press any key to continue . . . |' with a cursor at the end.

4.

Build a small script that takes the user's full name as input and creates a username by copying only the first 5 characters using strcpy(). Print the generated username.

Constraint: If the name is shorter than 5 characters, use the full name as the username.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
{
    char name[100];
    char username[6];

    printf("Enter your full name: ");
    fgets(name, sizeof(name), stdin);

    // Remove newline
    name[strcspn(name, "\n")] = '\0';

    if(strlen(name) <= 5)
    {
        strcpy(username, name);
    }
    else
    {
        strncpy(username, name, 5);
        username[5] = '\0';
    }

    printf("Generated Username: %s", username);

    return 0;
}
```



C:\Users\gajja\OneDrive\Desktop



Enter your full name: Manan Gajjar

Generated Username: Manan

Process exited after 3.443 seconds with return value 0

Press any key to continue . . . |